

Perfusion3D Hemodynamic & Coronary Ischemia Technical Evaluation

Model version: Perfusion3D v1.0.0

Date: 2026-10-05 11:56:25 UTC

Patient: 38 years, Male

1. Model output summary

Target	Vessel / condition	Probability	Model classification	Risk band
CAD	Overall Coronary Artery Disease	13.1%	Negative	● Low
LAD	Left Anterior Descending Artery	12.5%	Negative	● Low
LCX	Left Circumflex Artery	14.0%	Negative	● Low
RCA	Right Coronary Artery	13.8%	Negative	● Low

2. Clinical measurements

Demographics

Age	38 years	18-75
Sex	Male	Male/Female
Weight	66 kg	50-90
Length	166 cm	150-190
BMI	23.95 kg/m²	18.5-24.9

Clinical Examination

BP	110 mmHg	90-120
PR	70 bpm	60-100
DM	0	0
HTN	0	0
Typical Chest Pain	0	0

ECG

Q Wave	0	0
St Elevation	0	0
St Depression	0	0
Tinversion	0	0

Laboratory

FBS	80 mg/dL	70-99
CR	0.6 mg/dL	0.6-1.2
TG	41 mg/dL	50-150

LDL	85 mg/dL	50-100
HDL	65 mg/dL	40-60
Echocardiography		
EF-TTE	60 %	55-70
Region RWMA	0 score	0
VHD	N grade	N

3. Parameters outside typical reference range

No parameters outside reference range

4. Factors influencing model output

Target: CAD

Factor	Patient value	Effect on probability	SHAP value
cat__Typical Chest Pain_0	0	Lowers probability	-0.0738
cat__Typical Chest Pain_1	0	Lowers probability	-0.0726
num__Age	38.0 years	Lowers probability	-0.0361
cat__Atypical_N	N	Raises probability	0.0314
cat__Atypical_Y	N	Raises probability	0.0287

Target: LAD

Factor	Patient value	Effect on probability	SHAP value
cat__Typical Chest Pain_0	0	Lowers probability	-0.0867
cat__Typical Chest Pain_1	0	Lowers probability	-0.0673
num__EF-TTE	60.0 %	Lowers probability	-0.0379
num__Age	38.0 years	Lowers probability	-0.0353
cat__Region RWMA_0	0	Lowers probability	-0.0279

Target: LCX

Factor	Patient value	Effect on probability	SHAP value
num__Age	38.0 years	Lowers probability	-0.0737
cat__Typical Chest Pain_0	0	Lowers probability	-0.0451
cat__Typical Chest Pain_1	0	Lowers probability	-0.0407
num__TG	41.0 mg/dL	Lowers probability	-0.0302
num__HDL	65.0 mg/dL	Lowers probability	-0.0291

Target: RCA

Factor	Patient value	Effect on probability	SHAP value
cat__Typical Chest Pain_0	0	Lowers probability	-0.4125
num__Age	38.0 years	Lowers probability	-0.2481
num__TG	41.0 mg/dL	Lowers probability	-0.1705
num__HB	14.0 g/dL	Raises probability	0.1661
num__Neut	55.0 %	Lowers probability	-0.1557

5. Methodological notes

Evaluated using 55 non-invasive physiological features across 5 clinical categories.

Target features LAD, LCX, RCA, and Cath were strictly excluded from model inputs to prevent target leakage.

Reported outputs reflect calibrated probabilities derived from empirical post-test Bayesian odds, not direct anatomical lumen caliber measurements.

Predictions are for decision-support and educational purposes only and are not a substitute for formal diagnostic imaging or professional medical evaluation.